Science mock test: forces, pressure, digestion and breathing

Ember · 30 minutes · 33 marks · pen, pencil, ruler, calculator Date: _____

Calculations: write the equation, then the numbers, then the answer with its unit.
Explanations: one mark per link, so join your ideas with **so**.

[2 marks]

1. A box has a weight of 400 N. Its base has an area of 0.8 m². Calculate the pressure the box puts on the floor. Give the unit.

Working

Answer Pressure = _____ unit _____

[3 marks]

2. Explain why it hurts more to be stood on by someone in high-heeled shoes than in flat trainers, even though their weight is the same.

[3 marks]

3. A sealed balloon is put in a hot room. It gets bigger. Use ideas about particles to explain why.

[2 marks]

4. State what happens to atmospheric pressure as you climb a mountain, and give one reason why.

[5 marks]

5. Name the organs that food passes through, in order, from the mouth to the anus.

[3 marks]

6. Explain how the small intestine is adapted to absorb food quickly.

[2 marks]

7. Fibre is not digested. Explain why it is important to eat a fibre-rich diet.

[2 marks]

8. A spring is 5 cm long. A force of 3 N stretches it to 8 cm. (a) What is the extension? (b) The spring obeys Hooke's Law. What would the extension be with a 6 N force?

[3 marks]

9. A parachutist opens her parachute. Explain why she slows down and then falls at a steady speed.

[2 marks]

10. Give one way to reduce friction and one way to reduce drag.

[6 marks]

11. Explain how air is moved into and out of the lungs. Use the words volume and pressure in your answer.

Self-check (5 min)

□ Every calculation shows the equation and a unit.

□ Every explanation has its links joined with “so”.

□ Q11 covers breathing in **and** out, with volume **and** pressure.

□ I wrote down which questions I lost marks on.

Mark scheme

1. 400 ÷ 0.8 = **500** (1) **N/m²** or Pa (1).
2. Heels have a **smaller area** (1); the same force/weight on a smaller area (1) gives a **bigger pressure** (1).
3. The air particles get more energy and move **faster** (1); they hit the walls **harder / more often** (1); so the **pressure inside increases** and pushes the rubber out (1).
4. It **decreases** (1); there is less air above you / the air is less dense (1).
5. 1 mark each for **gullet, stomach, small intestine, large intestine, rectum**, in the right order between the mouth and the anus.
6. Villi give a **big surface area** (1); **thin wall**, so a short distance (1); **blood capillaries** carry the food away (1).
7. Fibre adds **bulk** (1) so the gut muscles can push food along / prevents **constipation** (1).
8. (a) 8 – 5 = **3 cm** (1). (b) double the force → double the extension: **6 cm** (1).
9. The open parachute has a big area, so **air resistance increases** (1); air resistance is bigger than her weight, so she **slows down** (1); as she slows, air resistance falls until it **equals her weight**: resultant force zero, so steady speed (1).
10. Friction: oil / grease / lubrication (1). Drag: streamlined shape (1).
11. In: rib muscles contract, ribs up and out (1); diaphragm contracts and moves down (1); chest volume increases and pressure decreases, so air is drawn in (1). Out: muscles relax, ribs down and in, diaphragm up (1); volume decreases (1); pressure increases, so air is pushed out (1).

Total: 33 marks. In the explain questions, a mark only counts if the *link* is there (for example “smaller area **so** bigger pressure”).